SOFTWARE ARCHITECTURE

Distributed Computing (cont.)

BEYOND BOUNDARIES

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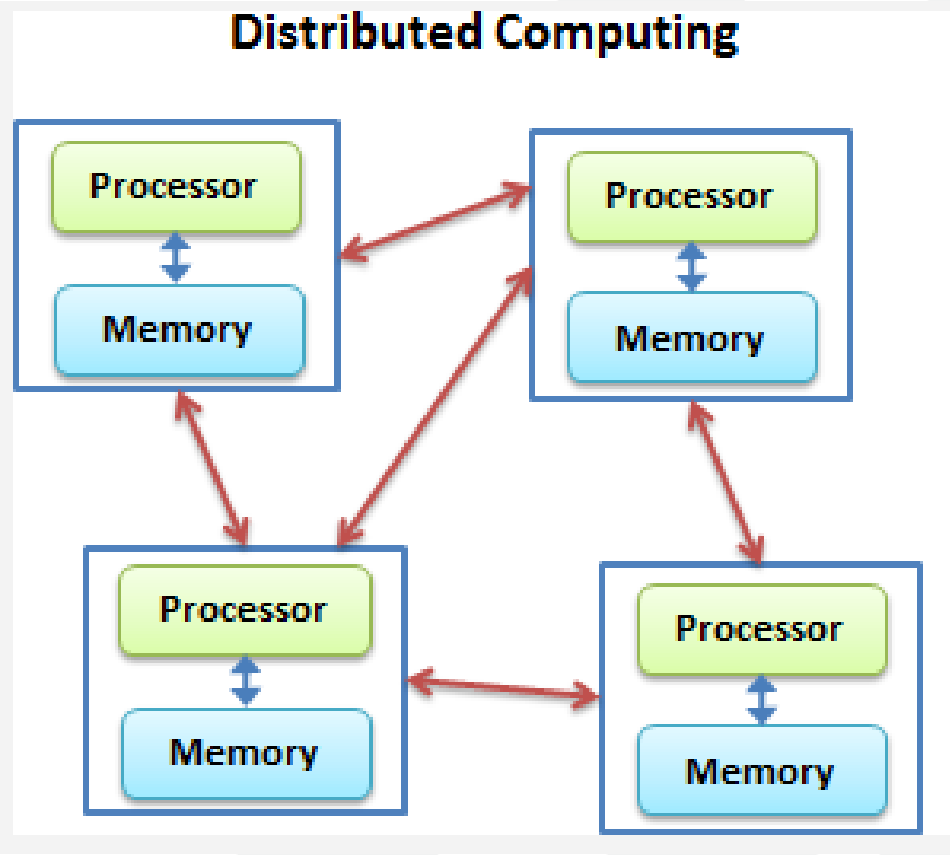
Problem statement

- it's too difficult to design distributed enterprise applications.
- With the loosely coupled system, the lack of latest current state and consistent information leads to many synchronization issues in the design of a distributed system.
- The resources of distributed applications are physically separated in their ecosystem.
- Due to the missing common resource factors, it is highly possible to have a delay in message delivery, and messages could even be lost.

Problem statement



The aims

- Distributed applications must be designed to leverage the complete advantages of the design to the end users.
- The distributed enterprise application can demonstrate system characteristics such as reliability, transparency, efficiency, security, and easiness.
- As a process to mitigate the listed challenges, the distributed applications are developed to deal with resolving several design issues.

Transparency

- Spatially separated multiple users consume the distributed applications concurrently.
- It is highly economical to build the sharing concept among the existing system resources for the concurrent execution of the user's request.
- By design, one of the existing processes must influence the other concurrently executing process as they might compete for the shared resources.
- As a result, collision might occur by making concurrent updates in the system. It is highly recommended that you avoid system collision.

Discussion

- Do you remember about ‘collision’ term? Let’s rethink about Data Structure and Algorithms course
- Discuss with your friend algorithms to solve this problem.

Concurrency transparency

- Concurrency transparency is a concept in which each user is intended to get a feeling of the sole usage of the resource in the system.
- To simulate the concept of concurrency transparency, the distributed system is recommended to get the following properties during the resource sharing processes:

Concurrency transparency

- Priority-based execution, it is proposed that you have an event-ordering property.
- To balance the deficiency against the need, no starvation property is established to ensure the granted resource by not using it simultaneously by multiple processes.
- To avoid conflict execution in the system, a third property is established, namely mutual-exclusion. This property ensures shared resource accessibility to be leveraged by one process at any point of time.
- To handle resource locking seamlessly, the final property is named no-deadlock property. The core purpose of this property is to prevent resource locking mutual progress while competing in the system.

Eight types of transparencies

- Access
- Migration
- Location
- Replication
- Relocation
- Concurrency
- Persistence
- Failure

Eight types of transparencies

- Access and Migration: difference in data representations and how ressource is accessed
- Location: where a resource is located
- Replication: share a resource by several competive users
- Relocation: move a resource to another location, while in use
- Concurrency: share resource by several competive users
- Persistence: whether a resource is in memory or on disk
- Failure: resources' failure and recovery process

Reliability

- A distributed system is expected to be more reliable than a centralized system.
- It seems that the existence of multiple instances increases the level of system reliability.
- On the flip side, few design challenges on loosely coupled systems have the probability of unreliable characteristics in distributed applications.
- Reliability is defined as the probability of success.

Fault tolerance

- In an enterprise distributed system, there is a high possibility of error generation due to any mechanical or algorithmic issue.
- Fault tolerance is defined as a system fault that leads to failure on executing the underlying applications.
- In general, faults are classified into three major categories. They are as follows:
  - Transient fault
  - Intermittent fault
  - Permanent fault

Fault tolerance

Transient faults occur once and then disappear, so it is **very difficult to reproduce** and resolve the bug.

Intermittent fault **repeats multiple** times with the characteristics of an occurring fault, and then it **disappears for a while, then it reoccurs**, and then it disappears, and so on.

Permanent fault has a persistent characteristic, so these types of fault **continue to exist until the faulty component is repaired or replaced**.

Performance

- Performance is quite an interesting field in the IT industry.
- In simple terms, it is the total number of tasks completed by a computer system.
- The core objective of the performance factor is to produce the output with a short response time for the given set of tasks.
- On comparison with a centralized system, distributed computing is expected to have better performance with high throughput.
- To meet the expectation of the preceding characteristics, it is essential to build some performance-oriented aspects in the distributed system.

Performance

Performance

Caching

Decompose

- In general, most of the complex large problems are solved in an easy way with the divide and conquer strategy.
- To illustrate the decomposition strategy, it is better to choose the common network use case. It is possible to transfer the data across the network in large chunks. Instead of transferring the complete large content in one shot, it is much more efficient to process the individual pages.

Caching

- In computing theory, some techniques, such as LRU (Least Recently Used), MRU (Most Recently Used), are quite common when it comes to improving performance using the concept of caching.
- The idea is to temporarily persist the frequently used data at the client site. In turn, the system is not expected to go and retrieve the data on every transaction. Instead, it can refer the cached data in the local layer.
- What is the benefit of doing this? Eventually, it improves overall system performance by saving a large amount of computing time and network latency.

Scalability

SCALE UP

SCALE OUT

COMPARING SCALE UP WITH SCALE OUT

CONNECTING THE DOTS

Scale up

- In terms of the scaling process, scale up is the traditional model to improve the power of the underlying system.
- As new resources are added in the same box, it is interpreted as vertical scaling. This is depicted in the following figure:

Scale up example

- A personal laptop
  - The initial hard disk configuration might be 500 GB.
  - After a while, the disk gets full and there is a pressing need to increase the volume of the hard disk.
  - So, the next step might be to increase the volume of the existing hard disk capacity to, say, 1 TB.
- Scale up is applicable to any type of system resources, such as RAM, processor, disk storage, application, and so on.

Scale out

- On similar lines, scale out is another methodology to improve the system performance of an enterprise application.
- The major difference is in the way of scaling.
- As demand increases, the additional systems are amended instead of adding the resources to the existing system. So, this design of scaling is termed scaling out or horizontal scaling. It is depicted in the following figure:

Comparing scale up with scale out

Sr.No.	Designfactor	Scale up	Scale out
1	Server node	Single server withhigh power.	Clustered set of server nodes with less power.2
	Computing	Concurrentcomputing.	Distributed computing.
3	Memory	High powered memory in a single computer.	Less powered memory but collectively simulate highpower memory.
4	Processor	Large count of processors in a single computer.	Less processors in a single computer but a cluster of computers is used.
5	Implementchallenges	Generally, less challenging to implement.	High degree of complexity toimplement.
6	Fault tolerance	High possibility offailure.	Easier to run fault tolerance.
7	Utility costs (electricity, cooling)	Low.	High.
8	Network bandwidth	Relatively less due of the single system design.	High degree of network bandwidth for distribution.
9	Vendor lock	Generally severe vendor lock-in due toproprietary	Open source leads to getting away from vendor lock.
10	Total Cost	High.	Relatively low.

Connecting the dots

- On connecting the dots of vertical scaling and horizontal scaling, here is the best use case.
- Generally, disk drives are a good analogy to the scale-up approach, whereas storage virtualization is a good analogy to the scale-out approach.
- On observing the recent emerging technology trends of the industry, disk capacity has increased significantly in alignment with processor capabilities, such as multicore technology.

Security

- Security is one of the most important principles in an enterprise application.
- As distributed computing is based on the cluster of the computers, it is highly recommended that you build a stronger security policy here.

Security

Goals

Attack

Threats

Goals

- In general, computer security has four goals:
  - Privacy
  - Secrecy
  - Authenticity
  - Integrity

Goals

- Privacy
  - The information to be used only for the intended purpose, not beyond that.
  - Example
    - In the financial industry, there is a term called PII (Personally Identifiable Information). It can be used on its own or with its supplementary information to identify the individual in context.
    - As per the US privacy law and information security, it is mandatory to protect this type of data.

Goals

- Secrecy
  - Is scoped to access only by authorized users, not anyone else.
  - Example
    - In an enterprise application, there might be an administration section to maintain the user base of the system.
    - This section is not supposed to be accessed by anyone in the system. Only some restricted users (such as the administrator) must have the accessibility to this section.

Goals

- Authenticity
  - Authenticity is defined as the ability to verify the received request only from the restricted users, not everyone.
  - Example
    - Private banking by end customers.
    - A bank might maintain the accounts of some n users.
    - The banking application can only be logged into by the registered n customers, not by other people.

Goals

- Integrity
  - Integrity is nothing but maintenance to ensure the accuracy and consistency of the information in an enterprise application.
  - At the same time, the information must be protected from unauthorized access.
  - With system characteristics, it is more integrated into the information reliability factor.

Attack

- An attack is any attempt to destroy, disable, or steal unauthorized system access.
- It violates the security discipline of the proper usage of the underlying system asset.

Attack

PASSIVE ATTACK

ACTIVE ATTACK

Attack

- Passive Attack:
  - Passive attack does not cause any harm to the system with its unauthorized access.
  - Example: change something on the UI
- Active Attack:
  - An attack creates a significant impact on the existing process and so it needs to be addressed thoroughly.
  - Example: The virus program, worm code.

Threats

- Threat is a possible security danger in the computing system. It might exploit a vulnerability to breach the security. So, it causes possible harm.
- There are four categories of threats:
  - Interception
  - Interruption
  - Fabrication
  - Modification

Threats

- Interception
  - Interception means gaining access to unauthorized service or data in a computer system. Illegal copying is an appropriate example.
- Interruption
  - Interruption makes the system resources unavailable or destroyed, for example, intentional file corruption.
- Fabrication
  - Fabrication is related to including the unwanted additional data.
- Modification
  - Modification alters the original specification of the system to break using unauthorized access.

Summary

- You were introduced to the fundamental concepts of distributed computing.
  - Multiprogramming
  - Multithreading
  - Design challenges
  - Scalability
  - Security

Q&A

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